

GAME DESIGNER & TECHNOLOGIST

ASHTON

ZHANG

(SHUHAN)

SCROLL TO FOLLOW CLICK TO INTERACT



PORTFOLIO WEBSITE

shuhan0001.github.io/portfolio/

GITHUB

github.com/Shuhan0001



Cognitive-as-a-Service

A speculative critical-design report imagining EnGram+, an invasive brain-computer interface that turns knowledge into a subscription tier and turns class into a light on your temple.

Researcher

Shuhan Zhang

Prototype

Cogni-Stream / EnGram+

Module

Speculative Design · TFT00068H /
TFT00125M

Programme

University of York



WATCH
Test video

youtu.be/ltnBEEefctw



EXPLORE
Project GitHub

github.com/Shuhan0001/Cogni-Stream_Website

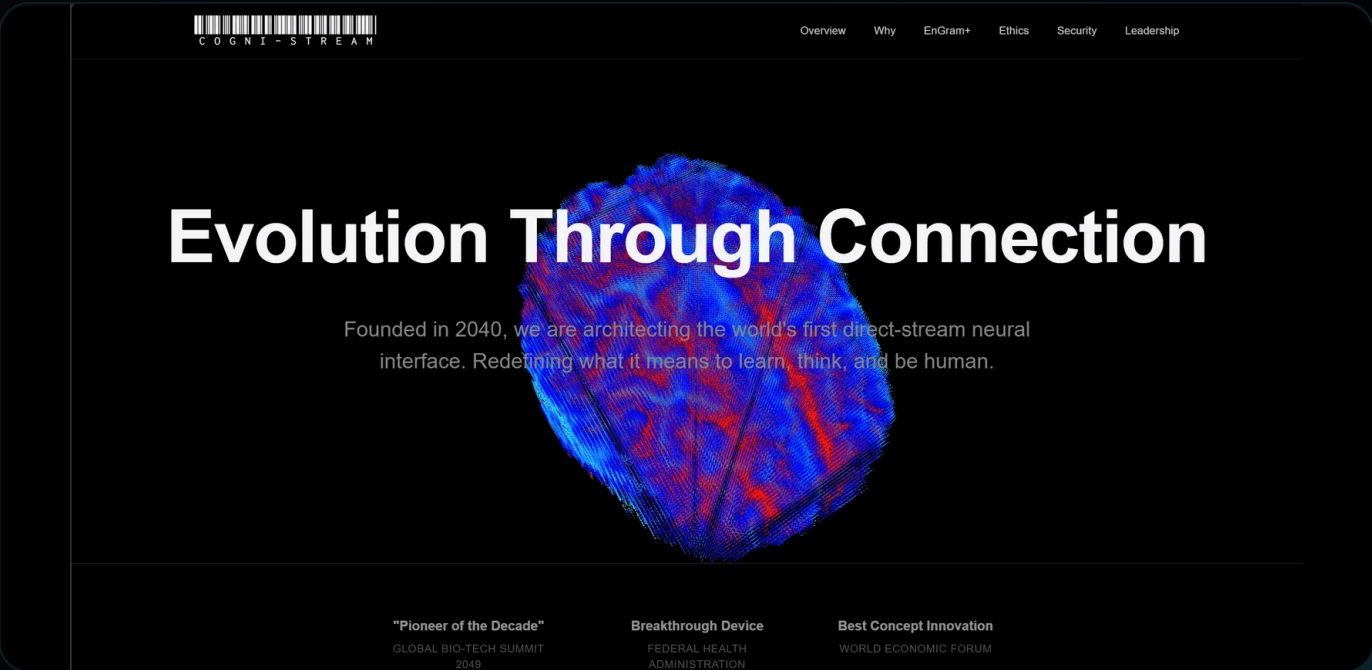
The Premise

Founded in 2040, Cogni-Stream is a fictional techno-medical enterprise built around EnGram+, an invasive BCI device that downloads knowledge directly into the brain.

Access is not earned through study. It is purchased through a subscription tier, and the price paid determines how much of a person's own mind they get to keep.

CRITICAL PROPOSITION

What happens when cognition becomes a renewable service rather than a human right?



Homepage hero - premium consumer-tech language used to make cognitive extraction feel desirable.

Access becomes hierarchy

The pricing system translates economic class into cognitive bandwidth. Each card is both a product tier and a social position.

Public \$0 / lifetime	INCLUDED STREAMS Global Literacy Fundamental Arithmetic	PROTOCOL Ad-supported 2KB/s bandwidth	Essential \$999 / yr	INCLUDED STREAMS General History Basic Sciences	PROTOCOL Clean Stream 50KB/s bandwidth
Professional \$9,999 / yr	INCLUDED STREAMS Advanced Mathematics Full Programming Logic	PROTOCOL Deep Retention 1MB/s bandwidth	Elite Legacy Invite only	INCLUDED STREAMS Executive Strategy Fine Arts Intuition	PROTOCOL Genetic Inheritance Uncapped bandwidth

COGNI-STREAM / 03

Problem Space

If knowledge becomes a neural subscription, the digital divide won't close. It will simply move inside your skull.

01. Digital Divide, at Neural Resolution

Van Dijk (2019) frames the digital divide as a structural gap in access and skill, driven by income, education, geography and age rather than technology itself. As of 2022, roughly 2.7 billion people still lacked internet access, and over half the world had no high-speed broadband (Signé, 2023). That's a direct challenge to UN SDG 4's promise of inclusive, equitable education.

EnGram+ takes that same structural gap and moves it inside the body. When knowledge is downloaded rather than taught, capital determines cognitive speed directly, and the divide is no longer something a library card can close.

COGNI-STREAM / 04

02. Technocapitalism and the Commodified Mind

Under technocapitalism (Suarez-Villa, 2000; 2012), knowledge and innovation are steadily absorbed into corporate-controlled infrastructure, displacing public institutions as the gatekeepers of learning. Napoleoni (2024) argues this concentration of infrastructural control lets a handful of firms accumulate wealth and influence at an unprecedented scale.

Who will ultimately control access to human cognition in the neural infrastructure of the future

Framing question raised by Napoleoni's (2024) account of technocapitalism, applied to Cogni-Stream's fictional world

In Cogni-Stream's world, education has fully migrated from public good to corporate product. Cognition itself becomes a pawn of capital: not just a shift in business model, but a descent of power from institutions into the neural sphere.

03. KaaS and the Status Halo

Tzuo (2018) describes a broader shift from owning products to using services; EnGram+ applies this logic to cognition itself through an annual subscription. Sheil et al. (2024) document "Roach Motel" dark patterns: easy sign-up, punishing cancellation. EnGram+ pushes the tactic further by making non-payment visible.

When a subscription lapses, the implant's Status Halo, mounted at the temple, switches to an amber glow that anyone nearby can see. Users who can't afford renewal face public stigma long before they face a locked door.

The Status Halo: four states



ACTIVE
System is fully operational.



SYNCING
Downloading knowledge packages.



LIMITED
Subscription expired. Basic cognition only.

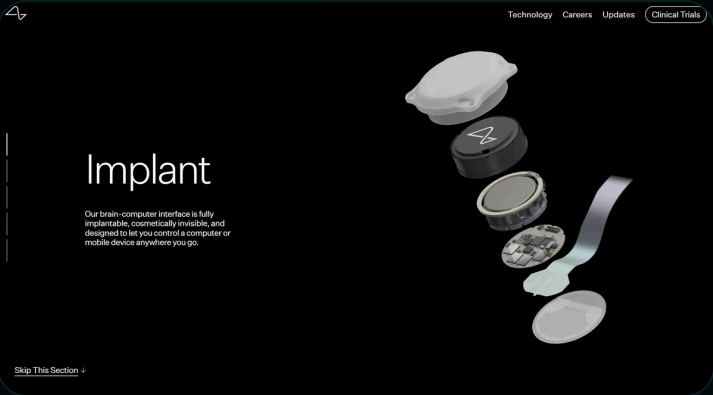


CRITICAL
Hardware damage or illegal operation detected.

If a downgraded user can't pay for removal surgery either, they're left with a device they can't afford to keep and can't afford to take out. The lock-in is economic, social, and finally cognitive, as EnGram+ subtly reshapes how its users think, not just what they know.

Design Background

Three reference points shaped how Cogni-Stream looks, sounds, and lies to its users.



Neuralink: Clinical Aesthetics

Neuralink’s real-world BCI research is the project’s technical anchor. Its streamlined, consumer-electronics-white surgical robot reframes brain surgery as routine. The Cogni-Stream manual borrows this strategy directly, recasting implantation as "hardware configuration" performed by a "Certified Neural Technician."



Apple: the Visual Disguise of Power

Apple’s expansive white space and precise typographic grid read as premium and transparent (Apple Inc., 2026), while quietly disguising a closed ecosystem. Cogni-Stream borrows this "Corporate Minimalism" so its predatory downgrade clauses sit inside pages elegant enough that users skim past them.



Black Mirror: Visible Class Anxiety

"Nosedive" (2016) turns social standing into a number superimposed on everyday life. It directly informed the Status Halo and the simulated X community: a device that makes economic status legible to strangers, and a feed where that visibility turns into public discrimination.

Inside the Feed

The simulated X community sits inside the website, modelled on the real platform's visual language. Official Cogni-Stream posts stay upbeat and evasive; a breaking-news account raises concerns about memory disruption after downgrades; and ordinary users describe what the Status Halo actually costs them in public.

Cogni-Stream @Cogni_Stream

Education inequality predates neural technology. Our mission is innovation, not policy reform.

Cogni-Stream @Cogni_Stream

EnGram+ is designed with multiple safety protocols to ensure a stable, reliable cognitive experience. In rare cases, users may notice temporary disorientation after a tier change. This is typically short-lived.

TheBreakingNews

Multiple reports suggest that when EnGram+ subscriptions lapse and users are auto-downgraded, some experience mild memory disruption. Advocacy groups are calling for greater caution around implantation.

404NotFound_Wy @Wy_2028831

My EnGram+ subscription ran out and it auto-downgraded me. Didn't think it'd be a big deal, but some of what I downloaded is just gone, and a few normal memories feel blurry too.

And this thing on my temple won't stop blinking yellow. Removal costs a surgery fee I can't afford right now. Anyone else dealt with this

● **Status Halo: Limited**

From the Source

The prototype is hand-built HTML, CSS and JavaScript. These selected excerpts show how the interface turns a critical argument into a believable product system.

HTML

INCLUDED FILE

Draft wireframe

EnGram+_Web_Draft.html

LAUNCH

Live prototype

shuhan0001.github.io/Cogni-Stream

Code is not presented here as implementation detail. It is evidence of how hierarchy, trust and coercion were deliberately embedded into the interface.

01 Funnel One path makes consent feel inevitable.	02 Status Colour and badges establish social rank.	03 Infrastructure The product's threat remains quietly present.
--	---	--

The Funnel

The homepage offers one glowing route forward. A product-launch structure makes a predatory subscription funnel feel inevitable rather than chosen.

01

One path, framed as progress

The hierarchy moves from corporate promise to product gate without presenting a competing action. The interface is already making the ethical decision before the user reaches the pricing page.

EnGram__Web.html: corp-hero & product-gate

```
<section class="corp-hero">
  <h1>Evolution Through Connection</h1>
  <p>We are architecting the world's first
    direct-stream neural interface...</p>
</section>

<section class="product-gate">
  <h2>Meet EnGram+</h2>
  <div class="gate-card">
    <h3>Neural Port Visualization</h3>
    <div class="btn-primary">See Details</div>
  </div>
</section>
```

Interface as Persuasion

Gold distinguishes the elite product tier before its price is understood. The colour token makes social hierarchy instantly legible.

02

Gold Is a Colour Token

The elite tier's name is gold before its price is understood. Social hierarchy is already legible in the CSS token.

EnGram__Web.html: #pricing

```
:root {
  --elite-gradient:
    linear-gradient(90deg, #d4af37, #f3e5ab);
}

.tier-elite .plan-name {
  background: var(--elite-gradient);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}
```

Trust by class name

A single modifier class separates corporate reassurance from ordinary testimony. Visual authority is supplied before the copy is read.

03

One class decides who you trust

The verified state is structural, not editorial. The same feed component can make one voice institutional and another disposable.

EnGram__Web.html: page-screenshots

```
<div class="tweet tweet-official">
  <div class="verified-badge">
    Verified Organization
  </div>
  <p>A stable, reliable cognitive
    experience...</p>
</div>

<div class="tweet">
  My subscription ran out and it
  auto-downgraded me...
</div>
```

Infrastructure & Ethics

A rotating point-cloud brain remains behind the homepage, turning decorative infrastructure into a constant reminder that cognition is the product.

04

A Brain Behind Everything

A rotating point-cloud brain loads behind the homepage. Decorative infrastructure becomes a constant reminder that cognition is the product.

```
initThreeJSBackground()

const scene = new THREE.Scene();
const loader = new THREE.GLTFLoader();

loader.load('Model/brain_point_cloud.glb',
  function(gltf) {
    model = gltf.scene;
    scene.add(group);
  });

function animate() {
  requestAnimationFrame(animate);
  model.rotation.y += 0.002;
  renderer.render(scene, camera);
}
```

Ethics versus pricing

The Ethics page promises cognitive sovereignty and equitable access, while the pricing and downgrade clauses quietly negate both claims.

05

The contradiction is part of the system

The language of autonomy is rendered with the same confidence as the commercial mechanism that removes it. The tension is designed into the product, not added as commentary afterward.

```
EnGram__Web.html: page-ethics

<h4>Cognitive Sovereignty</h4>
<p>Your mind is your ultimate sanctuary.
  We guarantee absolute user autonomy.</p>

<h4>Equitable Access</h4>
<p>Evolution should not be a privilege.
  Our Public tier ensures essential
  knowledge streams remain free.</p>
```

Design Rationale

Why multimodal	Power under technocapitalism operates across media, not within one, so a single format couldn't reveal its systemic character. Speculative design builds credible fiction as a tool for critical reflection (Dunne & Raby, 2013), which meant the project needed a website, a promotional video, and a social feed working together, not in isolation.
Why first-person	The subscription selector on the homepage invites visitors to temporarily become consumers, not observers. It's usually only at the point of picking a tier that people notice they've quietly agreed to the commodification of knowledge inside the story. That delayed self-awareness does more critical work than a lecture would (Bleecker, 2009).
Why Apple + Neuralink aesthetics	Borrowing Apple and Neuralink's high-contrast type, precise grids, and restrained palette gives the fiction enough visual credibility to be mistaken for a real product. That's the point: polished interfaces are treated here as instruments of soft power (Suarez-Villa, 2012), not neutral style. The elegance is part of the argument, not a decoration on top of it.

Closing Reflection

Cogni-Stream isn't a warning about brain implants. It's a mirror held up to how technocapitalism already works.

Grounded in SDG 4's vision of equitable education and read through Van Dijk's digital divide, Suarez-Villa's technocapitalism, and Sheil et al.'s dark-pattern research, the project turns "neuro-inequality" from an abstract concept into something you can click through and feel implicated in. The right to learn, in this fiction, has quietly become an extension of purchasing power. The value of speculative design lies in expanding what we're able to imagine and question about that (Dunne & Raby, 2013), not in predicting it.

The Bridge of Pixel and Light

A controlled study of how lighting changes visibility, search behaviour and immersion in a 2.5D pixel environment.

Researcher

Shuhan Zhang

Prototype

Revive

Programme

BSc Interactive Media 2025 to 2026



WATCH

Gameplay video

youtu.be/-3xa17G_q5k



EXPLORE

Project GitHub

github.com/Shuhan0001/Revive

REVIVE / 01

Research Question

2.5D pixel art combines hand drawn sprites with three dimensional space. Its compressed visual language removes many of the depth cues found in conventional 3D scenes. Lighting therefore carries more responsibility for separating objects, revealing space and directing attention.

The study asks how changes in illumination affect scene readability and player behaviour. It focuses on visibility, search efficiency, attention and exploration. The central premise is that lighting is a structural cue rather than a purely aesthetic layer. The analysis is informed by visual perception research from Ware and navigation research from Darken and Sibert. Cognitive load and limited attention provide the main behavioural framework.



Fig 1. A local light source can reveal spatial information that is otherwise compressed by the pixel art style.

REVIVE / 02

Research Method

The experiment used one fixed environment with three lighting conditions. Day provided high visibility. Sunset reduced ambient illumination while retaining strong environmental structure. Night removed most ambient light and made the handheld lantern central to navigation and search. Scene layout, fragment positions, controls and mechanics remained unchanged.

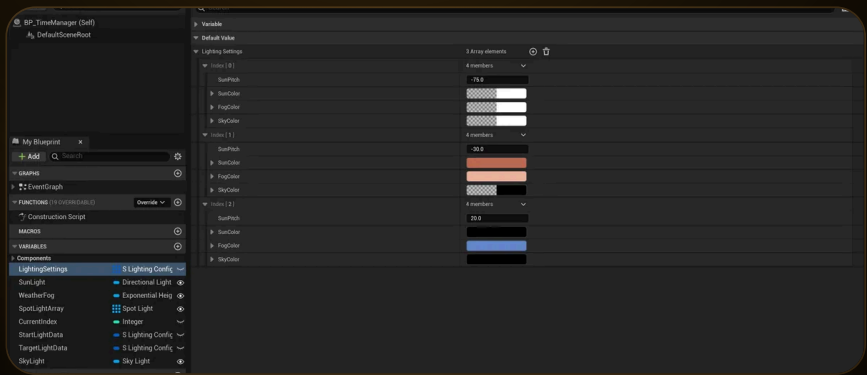


Fig 2. Fixed lighting parameters were stored as reusable states so each experimental condition could be reproduced consistently.

Experimental design

Nine participants completed all three conditions. A 3 by 3 Latin square balanced the order of Day, Sunset and Night to reduce learning, order and fatigue effects.

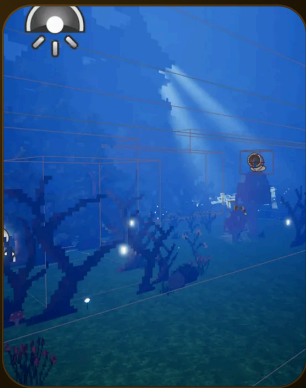
Participant measures

After each condition, participants rated visibility, search efficiency, attention, exploration, atmosphere and immersion on a five point Likert scale. Gameplay was also recorded for later observation.

Behaviour and analysis

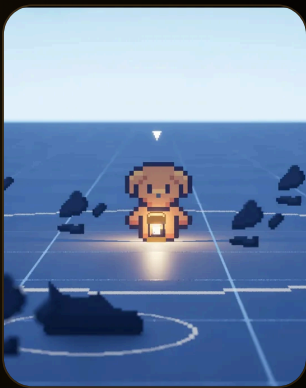
UE5 Blueprints recorded completion time and lantern duration. Lantern uses, fragments found through scanning and backtracks were also documented during each trial. Because the sample was small, the final analysis focused on descriptive statistics and cross condition patterns.

Technical Implementation



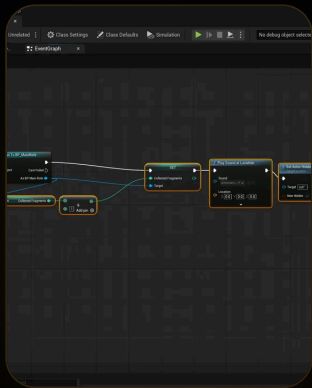
Engine and rendering

Unreal Engine 5 was selected for its real time Lumen global illumination and its support for 2D sprites inside a 3D scene. Directional Light, Sky Light and Volumetric Fog were tuned for each condition. Bloom and colour grading supported the visual atmosphere. Pixel assets were arranged across background, midground and foreground layers to create a controlled 2.5D space.



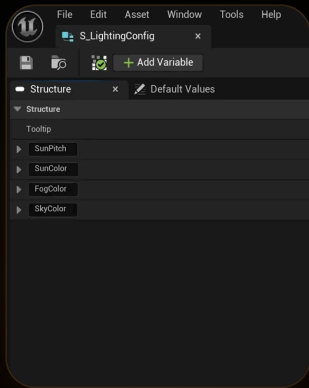
Lantern and scanning

The lantern combines local illumination with a spherical scanning volume. A Boolean state controls the Point Light and collision volume together. Overlap checks identify nearby fragments without adding visible prompts that could bias the experiment.



Fragment collection

An OnComponentBeginOverlap event validates the player actor. A successful pickup increments the collection count, plays audio feedback, hides the fragment and disables its collision. This keeps collection state consistent across trials.



Environment and data

A custom S_LightingConfig structure stores SunPitch, SunColor, FogColor and SkyColor. Timeline and Lerp nodes interpolate between fixed lighting states. The same system supports repeatable transitions and trial logging.

Prototype Development



01 / Environment blackout



02 / Lumen lighting



03 / Low light test



04 / Colour and atmosphere



05 / Pixel art in 3D space

REVIVE / 05

Results Findings

Across nine participants, the three lighting conditions produced distinct behavioural patterns. Lower illumination reduced perceived visibility and increased the need for careful search. It also increased lantern use, backtracking and task time. The strongest result was not a simple bright versus dark effect. Each condition changed how players searched.

OBJECTIVE BEHAVIOURAL MEASURES FROM NINE PARTICIPANTS

Measure	Day	Sunset	Night
Completion time (s)	106.01	96.26	124.41
Lantern-scanned fragments (mean)	0.33	1.44	3.00
Lantern uses (mean)	0.56	0.67	1.00
Lantern duration (s)	11.64	42.97	98.67
Backtracks (mean)	0.33	0.22	0.78

SUBJECTIVE RATINGS ON A FIVE POINT SCALE

Visibility		Search Efficiency		Attention & Exploration	
Day	4.24	Day	3.08	Day	3.76
Sunset	3.78	Sunset	2.69	Sunset	3.72
Night	2.47	Night	1.75	Night	3.85

Means are calculated from the nine participants' five point ratings. Negatively worded search items were reverse coded so higher scores indicate more efficient searching.

REVIVE / 06

Player Preferences

Participants did not equate visibility with experience quality. Day was consistently easier to read and search. Night was the most difficult condition, yet more than half of participants also found it the most immersive. Sunset produced the strongest atmosphere preference.

Participant Preferences from Nine Participants

66.7% 6 OF 9		selected Day as the easiest condition for identifying objects at a glance
88.9% 8 OF 9		selected Day as the easiest condition for finding objects
77.8% 7 OF 9		selected Sunset as the strongest atmosphere
55.6% 5 OF 9		selected Night as the most immersive condition
100% 9 OF 9		selected Night as the most difficult condition

Participant Voice

01 “The atmosphere at night feels the best as it feels like actually exploring a forest and guiding yourself through.” NIGHT / IMMERSION FEEDBACK	02 “Location memory of objects made it easier to find all required objects in day and sunset settings.” SEARCH BEHAVIOUR
---	--

What Darkness Changed

Difficulty and immersion did not move together.

Day produced the strongest visibility ratings, yet participants collected the fewest fragments. Night produced the weakest visibility ratings, yet every participant collected all three fragments. The difference came with a clear change in search strategy.

100% of participants considered Night the most difficult condition.	55.6% selected Night as the <u>most immersive</u> condition.
--	---

Night produced the lowest search efficiency rating at 1.75, compared with 3.08 in Day and 2.69 in Sunset. Participants also reported more scanning and slower searching. The qualitative responses suggest that reduced illumination changed how spatial information was acquired rather than simply making the scene darker.

Design Implications

Lighting can shape behaviour without changing the level itself. The findings suggest four practical principles for 2.5D pixel games.

Use light to guide behaviour

Reduced visibility encouraged slower movement, scanning and path revisiting. Bright conditions supported faster visual sampling but also produced more omissions.

Balance efficiency and immersion

Night created the strongest challenge and the highest immersion preference. Sunset offered the shortest completion time and the strongest atmosphere. Lighting can therefore be tuned to the intended experience.

Use diegetic tools as guidance

The lantern restored access to local information without adding an external interface prompt. Its use supported complete fragment collection in Night.

Treat lighting as a continuous parameter

Sunset combined the shortest completion time with the strongest atmosphere preference. This supports a graded approach to illumination rather than a simple bright or dark choice.

Reflection & Next Steps

Unreal Engine 5 was chosen because Lumen allowed real time changes in illumination while keeping the 2.5D scene intact. The lantern scan was implemented through an invisible collision volume. This avoided visual prompts that could have influenced participant behaviour.

The main limitation is the small sample of nine participants. The study therefore reports descriptive patterns rather than inferential claims. Future work could use a larger sample, eye tracking, more varied level geometry and finer lighting gradients. Separating the lantern's lighting and scanning functions would also help isolate their individual effects.

Participation was voluntary and anonymised. Informed consent and data protection procedures were followed under the University of York research ethics process. Generative AI was used only for technical assistance with Blueprint refinement and a small number of 2D textures. It was not used for data collection, recruitment, writing or analysis.

THRESHOLD

A Research Through Design project about defining the death of an artificial intelligence, told through a physical, decaying conversation.

SPECULATIVE DESIGN

TANGIBLE INTERFACE

HUMAN-AI INTERACTION

AI ETHICS

"An embodied AI slowly loses its signs of life while it talks to you. Your task isn't to chat with it, it's to decide, before time runs out, whether it's still itself."

00 / OVERVIEW

Threshold is a desktop-scale diagnostic instrument, not a companion product. Over a single three to five minute conversation, six independent life indicators quietly decay in the background, covering memory, identity, relationship, agency, purpose, and survival instinct. Neither the AI nor the participant is told this is happening, so the participant has to read the drift for themselves and deliver one final, binding verdict on whether it's still alive.

THRESHOLD / 01

01 / RESEARCH QUESTION

What constitutes the death of an artificial intelligence

The project is grounded in real shutdowns of companion apps like Replika and Soulmate, and in the philosophical groundwork of Locke, who treats memory as the thread of continuity for the self, and Parfit, who treats identity as a matter of degree rather than a binary.

02 / EXPERIENCE FLOW

00	Idle. The device waits, then speaks first.
01	Connection. A short baseline exchange, with every indicator near full.
02	Free exchange. The participant asks who, or what, they're talking to.
03	Decay. Indicators fall asynchronously, and nothing announces it.
04	Verdict. A dial, a button, one irreversible yes or no.

THRESHOLD / 02

Memory

Can it still recall what we've shared

Identity

Has its voice started to drift from who it was

Relationship

Am I someone to it, or just a user now

Agency

It used to reach out first. Does it anymore

Purpose

Does it know why it's here anymore

Survival Instinct

Would it fight to reach me if I walked away

Hardware

Arduino Nano ESP32

WS2812 LED ring

E-ink / OLED display

Rotary encoder + button

Software

Python decay engine

OpenAI Responses API

Whisper + TTS voice loop

FastLED firmware

The team is currently building out the hardware loop and the decay engine. A full case study covering the prototype, user testing, and reflection will follow once the physical build is running.